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Part I

Introduction

Part II

Theoretical Background

Chapter 1

Mori-Zwanzig Theory

The Mori-Zwanzig theory allows to observe the evolution of a dynamical system through the projection on a subset of its degrees of freedom, this is done through *projection operator techniques*.

The *projection operator technique* are powerful tools for the study of irreversible processes in many-body systems [7]. These tools has been introduced into statistical physics by S. Nakajima in 1958 and R. Zwanzig in 1960. The former to study the quantum statistical mechanics of steady diffusion [14], and the latter to give a new formulation of methods introduced in the statistical-mechanical theory of irreversibility [19]. In particular R. Zwanzig has developed the technique into a powerful tool for deriving formally accurate motion equations for classical or quantal probability densities [20, 21]. Later in 1965, H. Mori proposed a different tool, which leads to the linear *generalized Langevin equation* (GLE), an equation of the form

$$\frac{\mathrm{d}A}{\mathrm{d}t} = i\Omega A(t) - \int_0^t \mathrm{d}\tau K(\tau)A(t-\tau) + F(t) \tag{1.1}$$

where Ω, $K(t)$ and $F(t)$ are well-defined functions, called the frequency, memory function, and random force respectively. This equation describes the non-equilibrium evolution of any set of functions $A(t)$ defined on the phase-space of the arbitrary microscopic system, close to thermal equilibrium [13]. To now these variables $A(t)$ are called collective variables, relevant variables, coarse-grained (CG) variables or collective modes, depending on the context. Some years later in 1973 R. Zwanzig derived a class of generalized Langevin equations describing the motion of a system interacting with a heat bath, this GLE is exact but differs from that obtained by Mori's scheme for two reasons: for the presence of an arbitrary nonlinear force and for the fact that its validity is not restricted to small deviations from the thermal equilibrium [22]. In recent years D. Kauzlarić et all showed, in their work on the calculation of the friction matrix in the coarse-grained description of an atomistic system, that Mori GLE can be obtained from the Zwanzig's one if the system is close to the equilibrium and under the Markovian assumption, i.e. that the relevant variables (the functions $A(t)$ in the equation (1.1)) evolve in a time scale much larger than the memory time scale [10].

These techniques are know in various context as well as in transport theory. For example in the dynamic light scattering or in the liquids theory, the GLE

is used to compute time-correlation functions under circumstances where the usual phenomenological equations do not apply. Indeed through the equation (1.1) can be obtained an equation for the time correlation function $C(t)$:

$$\frac{\mathrm{d}C(t)}{\mathrm{d}t} = i\Omega\, C(t) - \int_0^t \mathrm{d}\tau K(\tau)C(t-\tau)\,. \tag{1.2}$$

This is called the *memory-function equation* and the random force appears implicitly in $K(t)$ [1, 8].

If the dynamical system considered is Hamiltonian, in the Mori-Zwanzig theory we can use two equivalent approaches. The former applying the projection operator in a Hilbert space of probability densities function (pdf) to the Liouville equation, obtained the generalized Fokker-Planck and master equations [11, 19, 20, 21]. While the latter, applying the projection operator in the Liouville space, see appendix C, to the equation of motion of time-dependent property phase function, obtained GLE and transport equation [3, 9, 13, 16, 22]. The mathematical equivalence of these two approaches is given by the duality between Liouville space and the space of densities of states, in complete analogy with the Schrödinger and Heisenberg pictures in Quantum Mechanics.

In what follows, we choose to follow the second approach since this results in a more physically-intuitive derivation and we will derive the GLE for the CG variables with the *Zwanzig projection* and the *Mori projection*, respectively.

We assume that at the microscopic level the system is well described by classical mechanics and is composed by M^{FG} particles. We will refer to it as the *fine-grained* (FG) system. The microscopic state is characterised by variable $\boldsymbol{\gamma}(t)$ in $6M^{FG}$-dimensional phase-space. To derive a reduced description of this system, we consider a smaller set of phase function $\boldsymbol{\Gamma}(\boldsymbol{\gamma})$, which we call observables[1]. The time evolution of a function of phase variables is given by

$$\frac{\mathrm{d}\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\,\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t))\,, \quad \boldsymbol{\Gamma}(\boldsymbol{\gamma}(0)) \equiv \boldsymbol{\Gamma}(\boldsymbol{\gamma}_0)\,. \tag{1.3}$$

Here, the operator $\mathfrak{L} \equiv iL$ is the *Liouvillian*, see appendix C, and the formal solution is given by

$$\boldsymbol{\Gamma}(\boldsymbol{\gamma}(t)) = \mathbf{e}^{\mathfrak{L}t}\,\boldsymbol{\Gamma}(\boldsymbol{\gamma}_0)\,, \tag{1.4}$$

where the exponential operator is defined through its Taylor expansion. This solution implicitly depends upon the full evolution of the FG system, therefore this equation does not provide a self-consistent evolution equation for the observables.

We also assume that the state of the system is described by the canonical equilibrium distribution[2] $\rho^{eq}(\boldsymbol{\gamma})$, and so that that the probability equilibrium measure is given by the Gibbs measure on the FG phase-space $X = \mathbb{R}^{6M^{FG}}$:

$$\mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \rho^{eq}(\boldsymbol{\gamma})\mathrm{d}\boldsymbol{\gamma} = \frac{1}{\mathcal{Z}}\mathbf{e}^{-\beta\mathcal{H}(\boldsymbol{\gamma})}\mathrm{d}\boldsymbol{\gamma}\,. \tag{1.5}$$

[1] Throughout the present section, these observables could be quite general, but we will later apply the derivation made here to the particular choice

$$\boldsymbol{\Gamma}(\boldsymbol{\gamma}) = (\boldsymbol{\Gamma}_{\mathbf{R}}(\boldsymbol{\gamma}), \boldsymbol{\Gamma}_{\mathbf{P}}(\boldsymbol{\gamma})) = (\mathbf{R}_1(\boldsymbol{\gamma}), \ldots, \mathbf{R}_{M^{CG}}(\boldsymbol{\gamma}), \mathbf{P}_1(\boldsymbol{\gamma}), \ldots, \mathbf{P}_{M^{CG}}(\boldsymbol{\gamma}))\,,$$

corresponding to the positions and momenta of particles in the CG system composed of M^{CG} particles.

[2] Different choices are possible. For example it is also perfectly possible to carry out the same derivation using the micro-canonical equilibrium measure.

The Mori-Zwanzig formalism allows us to replace equation (1.3) for the observables $\mathbf{\Gamma}(\boldsymbol{\gamma})$ by an equation of motion which depends on the microstate $\boldsymbol{\gamma}$ only through the CG variables themselves leading to a closed equation for the CG variables.

1.1 Mori approach

1.2 Zwanzig approach

1.2.1 The Zwanzig Projection

We denote the *Zwanzig projection* as $\mathcal{P}_{\mathbf{\Gamma}}$:

$$\mathcal{P}_{\mathbf{\Gamma}} F(\boldsymbol{\gamma}) = \frac{1}{\Omega(\mathbf{\Gamma}(\boldsymbol{\gamma}))} \int_X F(\boldsymbol{\gamma}') \, \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{\Gamma}(\boldsymbol{\gamma})\right) \rho^{eq}(\boldsymbol{\gamma}') \mathrm{d}\boldsymbol{\gamma}' \tag{1.6}$$

where $F(\boldsymbol{\gamma})$ is a generic function acting on the space-phase and the normalisation factor, also called the *structure function*, represents the total volume of the surface $S(\mathbf{\Gamma}(\boldsymbol{\gamma}))$ in phace-space, which is defined by $\mathbf{\Gamma}(\boldsymbol{\gamma}') = \mathbf{\Gamma}(\boldsymbol{\gamma})$:

$$\Omega(\mathbf{\Gamma}(\boldsymbol{\gamma})) = \int_X \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{\Gamma}(\boldsymbol{\gamma})\right) \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}') \,. \tag{1.7}$$

In other words, *structure function* represents the number of micro-states compatible with the macro-state $\mathbf{\Gamma}(\boldsymbol{\gamma}') = \mathbf{\Gamma}(\boldsymbol{\gamma})$ [9].

The projection operator $\mathcal{P}_{\mathbf{\Gamma}}$ selects from an arbitrary function $F(\boldsymbol{\gamma})$ that part, $\mathcal{P}_{\mathbf{\Gamma}} F(\boldsymbol{\gamma})$, which depends on $\boldsymbol{\gamma}$ only through $\mathbf{\Gamma}(\boldsymbol{\gamma})$. For a given $\boldsymbol{\gamma}$ we determine the numerical values $\mathbf{a}$ of observables, $\mathbf{a} = \mathbf{\Gamma}(\boldsymbol{\gamma})$. The integral in the numerator represents the amount of $F(\boldsymbol{\gamma})$ contained in the correspondence surface $S(\mathbf{a})$, while the *structure function* in the denominator $\Omega(\mathbf{\Gamma}(\boldsymbol{\gamma})) = \Omega(\mathbf{a})$ allows us to spread this amount on the surface $S(\mathbf{a})$. Therefore, loosely speaking, $\mathcal{P}_{\mathbf{\Gamma}}$ "average" the function $F(\boldsymbol{\gamma})$ over the surface $S(\mathbf{a})$. The fact that $\mathcal{P}_{\mathbf{\Gamma}}$ is a projection operator can be easily verified by showing that it satisfies the necessary and sufficient condition for a projection operator, $\mathcal{P}_{\mathbf{a}}^2 = \mathcal{P}_{\mathbf{a}}$ [20]. We can define the *complementary orthogonal projection* $\mathcal{Q}_{\mathbf{a}}$ as

$$\mathcal{Q}_{\mathbf{a}} = \mathcal{I} - \mathcal{P}_{\mathbf{a}} \,,$$

where $\mathcal{I}$ is the identity operator.

1.2.2 Zwanzig Projection as Conditional Expectation

We will now show that this projection is equivalent to taking the conditional expectation:

$$\mathcal{P}_{\mathbf{\Gamma}} F \equiv \mathbb{E}\left[F \middle| \mathbf{\Gamma}\right] = \langle F \rangle_{\mathbf{\Gamma}}$$

with respect to the equilibrium distribution, for a generic function $F = F(\boldsymbol{\gamma})$ acting on the phase space and for a specific value $\mathbf{a}$ of the CG variables $\mathbf{\Gamma}(\boldsymbol{\gamma})$ [2, 3]. This is done because it will be useful for the following sections.

Let A be a phase function in the Liouville space (see appendix C). We now focus our attention on a set of functions

$$\mathbf{A}(\boldsymbol{\gamma}) = \left\{ A_I \in L^2(X, \mathbb{P}) \,, \quad \text{where} \quad I = 1, \ldots, M^{CG} \quad \text{and} \quad M^{CG} \leq M^{FG} \right\} \,,$$

i.e. on the set of CG variables $\mathbf{\Gamma}$, which define a space $V = \mathbb{R}^{M^{CG}}$. Given a phase function B, we define the projection operator over the CG space V as

$$\mathcal{P}_{\mathbf{a}} B = \int_{A^{-1}(\{\mathbf{a}\})} B(\boldsymbol{\gamma}) \mathbb{P}_{\mathbf{a}}(\mathrm{d}\boldsymbol{\gamma}) \,,$$

where $A^{-1}(\{\mathbf{a}\}) = \{\boldsymbol{\gamma} \in X : \ \mathbf{A}(\boldsymbol{\gamma}) = \mathbf{a}\}$ is the preimage of $\mathbf{\Gamma} \in \mathbb{R}^{M^{CG}}$ for some fixed $\mathbf{a}$ and $\mathbb{P}_{\mathbf{a}}(\mathrm{d}\boldsymbol{\gamma})$ is the measure which satisfies

$$\int_V \left(\int_{A^{-1}(\{\mathbf{a}\})} B(\boldsymbol{\gamma}) \mathbb{P}_{\mathbf{a}}(\mathrm{d}\boldsymbol{\gamma}) \right) \mathrm{d}\mathbf{a} = \int_X B(\boldsymbol{\gamma}) \, \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) \,,$$

for any phase function B in the Liouville space.

Since that the observed value of the phase function $f(\boldsymbol{\gamma})$ is the expected value $\mathbb{E}[f]$, see definition A.9, with respect to the Gibbs measure (1.5), we can interpret the phase function as random variable. Therefore, the projection operator has a nice interpretation as a conditional expectation over the CG variables:

$$\mathcal{P}_{\mathbf{a}} F = \mathbb{E}\left[F \middle| \, \mathbf{A}(\boldsymbol{\gamma}) = \mathbf{a}\right] . \tag{1.8}$$

For the mathematical details of the conditional expectation see theorem A.11. We stress here that the conditional expectation 1.8, which we rewrite as $\mathbb{E}\left[B \middle| \, \mathbf{a}\right]$, is a function of $\mathbf{a}$, is a linear operator and represents the best approximation of B compared to any function of $\mathbf{a}$, is therefore an orthogonal projection. For further details on the last property, see the definition A.12.

1.2.3 Generalized Langevin Equation

We are now in the position to derive the GLE. We will make the derivation considering an ensemble of initial conditions $\boldsymbol{\gamma}_0$ satisfying $\mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{a}$ for some fixed $\mathbf{a} = \mathbf{A}(\boldsymbol{\gamma})$ and with $\boldsymbol{\gamma}_0$ distributed according to the equilibrium density $\rho^{eq}(\boldsymbol{\gamma}_0)$ conditional on $\mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{a}$.

Differentiating both sides of the relation (1.4) with respect to time, we can rewrite the equation (1.3) as

$$\frac{\mathrm{d}\mathbf{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\, \mathbf{e}^{\mathfrak{L}t}\, \mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{e}^{\mathfrak{L}t} \mathfrak{L}\, \mathbf{\Gamma}_0 \,, \tag{1.9}$$

where $\mathbf{\Gamma}_0 = \mathbf{\Gamma}(\boldsymbol{\gamma}_0)$. Applying the identity operator $\mathcal{I} = \mathcal{P} + \mathcal{Q}$ to the equation obtained, one has

$$\frac{\mathrm{d}\mathbf{\Gamma}_t}{\mathrm{d}t} = \mathbf{e}^{\mathfrak{L}t} \left(\mathcal{P} + \mathcal{Q}\right) \mathfrak{L}\, \mathbf{\Gamma}_0 = \mathbf{e}^{\mathfrak{L}t} \mathcal{P} \mathfrak{L}\, \mathbf{\Gamma}_0 + \mathbf{e}^{\mathfrak{L}t} \mathcal{Q} \mathfrak{L}\, \mathbf{\Gamma}_0 \,. \tag{1.10}$$

Now if we make in the Dunhamel-Dyson operator identity for the exponential operator $exp(-(A+B)t)$, see appendix D, the substitutions $A = \mathfrak{L}\mathcal{Q}$ and $B = \mathfrak{L}\mathcal{P}$, we get

$$\mathbf{e}^{-\mathfrak{L}t} = \mathbf{e}^{-\mathfrak{L}\mathcal{Q}t} - \int_0^t \mathbf{e}^{-\mathfrak{L}\mathcal{Q}s} \, \mathfrak{L}\mathcal{P} \, \mathbf{e}^{-\mathfrak{L}(t-s)} \, \mathrm{d}s \,,$$

and if we take the adjoint of both side, we obtain that

$$\mathbf{e}^{\mathfrak{L}t} = \mathbf{e}^{\mathcal{Q}\mathfrak{L}t} + \int_0^t \mathbf{e}^{\mathfrak{L}(t-s)} \, \mathcal{P}\mathfrak{L} \, \mathbf{e}^{\mathcal{Q}\mathfrak{L}s} \, \mathrm{d}s \,, \tag{1.11}$$

where we used the fact that the projection operators $\mathcal{P}$ and $\mathcal{Q}$ are Hermitian operator, while the liouvillian $\mathfrak{L}$ is anti-Hermitian.

We can rewrite the second term on the RHS of eq. (1.10) as

$$\mathbf{e}^{\mathfrak{L}t}\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0 = \mathbf{e}^{\mathcal{Q}\mathfrak{L}t}\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0 + \int_0^t \mathbf{e}^{\mathfrak{L}(t-s)}\,\mathcal{P}\mathfrak{L}\,\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0\,\mathrm{d}s\,, \tag{1.12}$$

and, considering the projection on the CG variables such that $A(\boldsymbol{\gamma}) = \mathbf{a}$, the last part of the integral becomes:

$$\begin{aligned}\mathcal{P}\mathfrak{L}\,\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0 &= \frac{1}{\Omega(\mathbf{a})}\int \mathrm{d}\boldsymbol{\gamma}_0\rho^{eq}(\boldsymbol{\gamma}_0)\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}_0)-\mathbf{a}\right)\left[\mathfrak{L}\,\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right] = \\ &= -\frac{1}{\Omega(\mathbf{a})}\int \mathrm{d}\boldsymbol{\gamma}_0\rho^{eq}(\boldsymbol{\gamma}_0)\left[\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right]\mathfrak{L}\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}_0)-\mathbf{a}\right)\,.\end{aligned}$$

Here, we used the fact that $\mathfrak{L}$ is skew-symmetric (anti-Hermitian) on the Liouville space, i.e. $(f, \mathfrak{L}\,g) = -(\mathfrak{L}\,f, g)$ for all f, g in the Liouville space. Using the chain rule for time differentiation[3] we can write

$$\mathfrak{L}\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma})-\mathbf{a}\right) = \mathfrak{L}\,\mathbf{\Gamma}\cdot\nabla_{\mathbf{\Gamma}}\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma})-\mathbf{a}\right) = -\mathfrak{L}\,\mathbf{\Gamma}\cdot\nabla_{\mathbf{a}}\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma})-\mathbf{a}\right)$$

where $\nabla_{\mathbf{x}}$ denotes the gradient taken with respect to the variable $\mathbf{x}$, and get:

$$\begin{aligned}\mathcal{P}\mathfrak{L}\,\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0 &= \frac{1}{\Omega(\mathbf{a})}\int \mathrm{d}\boldsymbol{\gamma}_0\rho^{eq}(\boldsymbol{\gamma}_0)\,\left[\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right]\mathfrak{L}\,\mathbf{\Gamma}_0\cdot\nabla_{\mathbf{a}}\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}_0)-\mathbf{a}\right) = \\ &= \frac{1}{\Omega(\mathbf{a})}\,\nabla_{\mathbf{a}}\cdot\int \mathrm{d}\boldsymbol{\gamma}_0\rho^{eq}(\boldsymbol{\gamma}_0)\,\left\{\,\delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}_0)-\mathbf{a}\right)\left[\left(\mathfrak{L}\,\mathbf{\Gamma}_0\right)\otimes\left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right)\right]\right\}\,,\end{aligned}$$

where $\otimes$ represents a tensor product. We used the fact that the divergence of a continuously differentiable second-order tensor field is a first-order tensor field, i.e. $\nabla\cdot(f\,\mathbf{u}\otimes\mathbf{v}) = \mathbf{v}\,(\mathbf{u}\cdot\nabla f) + f\,(\nabla\cdot\mathbf{u})\,\mathbf{v} + f\,(\mathbf{u}\cdot\nabla)\,\mathbf{v}$ where f is a generic function and $\mathbf{u}$, $\mathbf{v}$ two generic vectors. We can rewrite the equation obtained as

$$\begin{aligned}\mathcal{P}\mathfrak{L}\,\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\,\mathbf{\Gamma}_0 &= \frac{1}{\Omega(\mathbf{a})}\,\nabla_{\mathbf{a}}\cdot\left\{\Omega(\mathbf{a})\,\mathcal{P}\left[\left(\mathfrak{L}\,\mathbf{\Gamma}_0\right)\otimes\left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right)\right]\right\} = \\ &= \frac{\nabla_{\mathbf{a}}\Omega(\mathbf{a})}{\Omega(\mathbf{a})}\,\mathcal{P}\left[\left(\mathfrak{L}\,\mathbf{\Gamma}_0\right)\otimes\left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right)\right] + \nabla_{\mathbf{a}}\cdot\mathcal{P}\left[\left(\mathfrak{L}\,\mathbf{\Gamma}_0\right)\otimes\left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\,\mathcal{Q}\mathfrak{L}\mathbf{\Gamma}_0\right)\right] = \\ &= -\nabla_{\mathbf{a}}\,\mathcal{S}\,(\mathbf{a})\,\mathcal{M}\,(\mathbf{a},s) + \frac{1}{\beta}\,\nabla_{\mathbf{a}}\cdot\mathcal{M}\,(\mathbf{a},s)\,,\end{aligned} \tag{1.13}$$

where $\beta = (k_B T)^{-1}$ and we have introduced two new functions. The former

$$\mathcal{S}\,(\mathbf{a}) \coloneqq -\frac{1}{\beta}\ln\Omega\,(\mathbf{a}) \tag{1.14}$$

is interpreted, since it has units of energy, as a contribution to the free energy of the system, in view of the definition of $\Omega(\mathbf{a})$ in eq. (1.7). We note that in [9],

[3] For a generic function on the CG space:

$$\frac{\mathrm{d}B(\mathbf{\Gamma})}{\mathrm{d}t} = \mathfrak{L}\,B(\mathbf{\Gamma}) = \dot{\mathbf{\Gamma}}\cdot\nabla_{\mathbf{\Gamma}}B(\mathbf{\Gamma}) = \mathfrak{L}\,\mathbf{\Gamma}\cdot\nabla_{\mathbf{\Gamma}}B(\mathbf{\Gamma})\,,$$

so we get $\mathfrak{L}\,B(\mathbf{\Gamma}) = \mathfrak{L}\,\mathbf{\Gamma}\cdot\nabla_{\mathbf{\Gamma}}B(\mathbf{\Gamma})$.

the alternative quantity $\mathcal{S}(\mathbf{a}) = k_B \ln \Omega(\mathbf{a})$ is defined, which is an entropy. Both definitions are identical up to a multiplicative factor, and depend upon on the assumed temperature of the system. Instead, the latter function is a positive definite memory matrix:

$$\begin{aligned}\mathcal{M}(\mathbf{a}, s) &\coloneqq \beta \mathcal{P}\left[(\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right] = \\ &= \beta \left\{\mathcal{P}\left[(\mathcal{P}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right] + \mathcal{P}\left[(\mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right]\right\} = \\ &= \beta \left\{(\mathcal{P}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \mathcal{P}\left[\left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right] + \mathcal{P}\left[(\mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right]\right\} = \\ &= \beta \mathcal{P}\left[(\mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}\mathfrak{L}s}\, \mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right],\end{aligned} \tag{1.15}$$

where we used the fact that $\mathcal{P}\mathcal{Q} = 0$.

Inserting eq. (1.13) in eq. (1.12), the eq. (1.10) becomes

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{\Gamma}_t}{\mathrm{d}t} &= \mathbf{e}^{\mathfrak{L}t}\mathcal{P}\mathfrak{L}\boldsymbol{\Gamma}_0 + \mathbf{e}^{\mathfrak{L}t}\mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0 = \\ &= \mathbf{e}^{\mathfrak{L}t}\mathcal{P}\mathfrak{L}\boldsymbol{\Gamma}_0 - \int_0^t \mathbf{e}^{\mathfrak{L}(t-s)}\, \nabla_{\mathbf{a}}\mathcal{S}(\mathbf{a})\, \mathcal{M}(\mathbf{a}, s)\, \mathrm{d}s + \\ &\quad + \frac{1}{\beta}\int_0^t \mathbf{e}^{\mathfrak{L}(t-s)}\, \nabla_{\mathbf{a}} \cdot \mathcal{M}(\mathbf{a}, s)\, \mathrm{d}s + \mathbf{e}^{\mathcal{Q}\mathfrak{L}t}\mathcal{Q}\mathfrak{L}\boldsymbol{\Gamma}_0\,.\end{aligned} \tag{1.16}$$

Now, since for any $f(\boldsymbol{\Gamma}_t)$ we have $\mathbf{e}^{\mathfrak{L}t} f(\boldsymbol{\Gamma}_0) = f(\boldsymbol{\Gamma}_t)$, the terms in equation become:

$$\begin{aligned}\mathbf{e}^{\mathfrak{L}t}\mathcal{P}_{\mathbf{a}}\mathfrak{L}\boldsymbol{\Gamma}_0 &= \mathcal{P}_{\mathbf{a}(t)}\mathfrak{L}\boldsymbol{\Gamma}_0\,, \\ \mathbf{e}^{\mathfrak{L}(t-s)}\, \nabla_{\mathbf{a}}\mathcal{S}(\mathbf{a})\, \mathcal{M}(\mathbf{a}, s) &= \nabla_{\mathbf{a}}\mathcal{S}(\mathbf{a}(t-s))\, \mathcal{M}(\mathbf{a}(t-s), s)\,, \\ \mathbf{e}^{\mathfrak{L}(t-s)}\, \nabla_{\mathbf{a}} \cdot \mathcal{M}(\mathbf{a}, s) &= \nabla_{\mathbf{a}} \cdot \mathcal{M}(\mathbf{a}(t-s), s)\,,\end{aligned}$$

so we get

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{\Gamma}_t}{\mathrm{d}t} &= \Lambda(\boldsymbol{\Gamma}_t) - \int_0^t \nabla_{\mathbf{a}}\mathcal{S}(\mathbf{a}(t-s))\, \mathcal{M}(\mathbf{a}(t-s), s)\, \mathrm{d}s + \\ &\quad + \frac{1}{\beta}\int_0^t \nabla_{\mathbf{a}} \cdot \mathcal{M}(\mathbf{a}(t-s), s)\, \mathrm{d}s + \boldsymbol{\mathcal{F}}_{\boldsymbol{\Gamma}}(t, \boldsymbol{\gamma}_0)\,,\end{aligned} \tag{1.17}$$

where we have defined

$$\Lambda(\boldsymbol{\Gamma}_t) \coloneqq \mathcal{P}_{\mathbf{a}(t)}\mathfrak{L}\boldsymbol{\Gamma}_0\,, \tag{1.18}$$

$$\boldsymbol{\mathcal{F}}_{\boldsymbol{\Gamma}}(t, \boldsymbol{\gamma}_0) \coloneqq \mathbf{e}^{\mathcal{Q}_{\mathbf{a}}\mathfrak{L}t}\mathcal{Q}_{\mathbf{a}}\mathfrak{L}\boldsymbol{\Gamma}_0 = \mathcal{Q}_{\mathbf{a}}\mathbf{e}^{\mathcal{Q}_{\mathbf{a}}\mathfrak{L}t}\mathfrak{L}\boldsymbol{\Gamma}_0\,. \tag{1.19}$$

The quantity $\boldsymbol{\mathcal{F}}_{\boldsymbol{\Gamma}}(t, \boldsymbol{\gamma}_0)$ still depends on the FG variables, and is given by the solution of an auxiliary set of equations called *orthogonal dynamics equations* [6], which we will discuss in the following sections. We can now rewrite eq. (1.15) as

$$\begin{aligned}\mathcal{M}(\mathbf{a}, s) &\coloneqq \beta \mathcal{P}_{\mathbf{a}}\left[(\mathcal{Q}_{\mathbf{a}}\mathfrak{L}\boldsymbol{\Gamma}_0) \otimes \left(\mathbf{e}^{\mathcal{Q}_{\mathbf{a}}\mathfrak{L}s}\, \mathcal{Q}_{\mathbf{a}}\mathfrak{L}\boldsymbol{\Gamma}_0\right)\right] = \\ &= \beta\mathcal{P}_{\mathbf{a}}\left[\boldsymbol{\mathcal{F}}_{\boldsymbol{\Gamma}}(0, \boldsymbol{\gamma}_0) \otimes \boldsymbol{\mathcal{F}}_{\boldsymbol{\Gamma}}(t, \boldsymbol{\gamma}_0)\right],\end{aligned} \tag{1.20}$$

$$\mathcal{M}(\mathbf{a}, s) \coloneqq \beta \mathcal{P}_{\mathbf{a}} \left[(\mathcal{Q}_{\mathbf{a}} \mathfrak{L} \mathbf{\Gamma}_0) \otimes \left(\mathrm{e}^{\mathcal{Q}_{\mathbf{a}} \mathfrak{L} s} \mathcal{Q}_{\mathbf{a}} \mathfrak{L} \mathbf{\Gamma}_0 \right) \right]$$
$$\equiv \beta \mathbb{E} \left[\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(0, \boldsymbol{\gamma}_0) \otimes \boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0) | \, \mathbf{a} \right] \equiv \beta \left\langle \boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(0, \boldsymbol{\gamma}_0) \otimes \boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0) \right\rangle_{\mathbf{a}}$$

therefore, from eq. (1.17), $\boldsymbol{\mathcal{M}}$ is a friction matrix, whose components can be written in Green-Kubo form [10].

Recall that we are not interested in solving the equation (1.17) for a specific initial condition $\mathbf{\Gamma}(\boldsymbol{\gamma}_0)$, but rather for an ensemble of initial conditions $\boldsymbol{\gamma}_0$ satisfying $\mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{a}(0)$ for some fixed $\mathbf{a}(0) = \mathbf{A}(\boldsymbol{\gamma})$, where $\mathbf{A}(\boldsymbol{\gamma})$ is a generic set of CG variables, and with $\boldsymbol{\gamma}_0$ distributed according to the equilibrium density $\rho^{eq}(\boldsymbol{\gamma}_0)$ conditional on $\mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{a}(0)$. It is important to highlight here that the complexity of the equation (1.17) is not reduced with respect the starting equation (1.3), it has simply been shifted into the calculation of the *orthogonal dynamics*. However close to equilibrium, the term $\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0)$ is uncorrelated with the projected variables $\mathbf{\Gamma}$ and therefore approximated as a random noise. The proof of this statement is based on the scalar product defined in the Liouville space and the Mori projector $\mathcal{P}^M$, see section 1.1, which, as we said, is a special case of the Zwanzig projector when the system is near the equilibrium [10]. If we now consider eq. (1.19), we can write:

$$\begin{aligned}
(\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0), \mathbf{\Gamma}^*(\boldsymbol{\gamma}_0)) &= \left(\mathrm{e}^{\mathcal{Q}^M_{\mathbf{a}} \mathfrak{L} t} \mathcal{Q}^M_{\mathbf{a}} \mathfrak{L} \mathbf{\Gamma}_0, \mathbf{\Gamma}^*_0 \right) = \\
&= \left(\mathcal{Q}^M_{\mathbf{a}} \mathrm{e}^{\mathcal{Q}^M_{\mathbf{a}} \mathfrak{L} t} \mathcal{Q}^M_{\mathbf{a}} \mathfrak{L} \mathbf{\Gamma}_0, \mathbf{\Gamma}^*_0 \right) = \\
&= \left(\mathcal{Q}^M_{\mathbf{a}} \boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0), \mathbf{\Gamma}^*_0 \right) = \\
&= (\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0), \mathbf{\Gamma}^*_0) - \frac{(\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0), \mathbf{\Gamma}^*_0)}{(\mathbf{\Gamma}_0, \mathbf{\Gamma}^*_0)} (\mathbf{\Gamma}_0, \mathbf{\Gamma}^*_0) = 0 \,,
\end{aligned}$$

where we used the idempotent property of the projector and the definition of $\mathcal{Q}^M$, the complement of $\mathcal{P}^M$. In the Langevin description of the dynamics of a system, the random force has the property of being uncorrelated from the velocity of the particle. For the same reason, being the dynamic of $\boldsymbol{\mathcal{F}}_{\mathbf{\Gamma}}(t, \boldsymbol{\gamma}_0)$ orthogonal to that of $\mathbf{\Gamma}$, i.e. they are uncorrelated, we can identity it as random force acting on the system.

With this in mind, we can rewrite eq. (1.17) as an integro-differential equation with a random-source term which is usually referred to as the *Generalized Langevin Equation* (GLE):

$$\frac{\mathrm{d}\mathbf{\Gamma}_t}{\mathrm{d}t} = \Lambda(\mathbf{\Gamma}_t) - \int_0^t \nabla_{\mathbf{a}} \mathcal{S}(\mathbf{a}(t-s)) \, \mathcal{M}(\mathbf{a}(t-s), s) \, \mathrm{d}s + \\ + \frac{1}{\beta} \int_0^t \nabla_{\mathbf{a}} \cdot \mathcal{M}(\mathbf{a}(t-s), s) \, \mathrm{d}s + \mathbf{F}(t) \,, \quad (1.21)$$

where $\mathbf{F}(t)$ is interpreted as a zero-mean random process whose statistical properties are specified by eq. (1.19) in which $\boldsymbol{\gamma}_0$ is random and distributed according to the equilibrium density $\rho^{eq}(\boldsymbol{\gamma}_0)$ conditional on $\mathbf{\Gamma}(\boldsymbol{\gamma}_0) = \mathbf{a}(0)$.

The terms involving the friction matrix turn out to be the most difficult to compute in a real calculation because of the memory effect. We will briefly discuss the friction matrix and the memory effects but a more thorough analysis of these terms is left for future works. In the next chapter we will derive more explicit forms of the terms appearing in eq. (1.17) for a CG system, for specific choices of the CG variables which arises naturally from physical considerations, namely canonical mappings (i.e. transformations).

Chapter 2

Canonical Mapping and Coarse-Graining

2.1 Definition and Notation

In the present work we have said that we consider two resolution levels for a given molecular system. The first one, at high resolution, is the fine-grained (FG) system composed by M^{FG} particles, while the second one, at low resolution, is the CG system composed by M^{CG} particles. The term *particles* is used to indicate either an "atom", the fundamental particles that interacts in the FG system, or a larger group of "atoms", called *bead*, that interacts in the CG system.

We will assume that the chemical structure of the molecular system is fixed, i.e. no reactions occur, the connectivity of the "atoms" is specified at the outset allowing us to fix a global index for the particles in the FG system, and that each "atom" is mapped to a fixed (unique) particle in the CG system. Following [15], we can summarise the notation and the definition that we will use, and that we have already used, in the following way:

- lower case letters (e.g. i, j) refer to particles in the FG system, while capital letters (e.g. I, J) label the particles ("atoms" or beads) in the CG system;
- the mass of the j-th atom in FG system will be indicated with a lower case m as m_j, while the mass of the J-th particle in CG system will be indicated as M_J;
- the set of "atoms" included within the bead of index J is define as

  $$S_J \coloneqq \left\{ j \in \left\{1, \dots, M^{FG}\right\} : \text{the } j\text{-th atom forms part of CG particle } J \right\} .$$

  $$s_I = \#\left\{ i \in \left\{1, \dots, M^{FG}\right\} : \text{the } i\text{-th atom forms part of CG particle } I \right\} .$$

- the number of "atoms" corresponding to CG particle J will be referred to as $s_J = \#S_J$;
- the index of the CG particle containing the k-th "atom" will be referred to as J_k;

- FG phase space variables are denoted with lower case letters:

$$\gamma = (\gamma_{\mathbf{r}}, \gamma_{\mathbf{p}}) = (\mathbf{r}_1, \ldots, \mathbf{r}_{M^{FG}}, \mathbf{p}_1, \ldots, \mathbf{p}_{M^{FG}}) \ ;$$

- CG phase space variables are denoted using capital letters:

$$\Gamma = (\Gamma_{\mathbf{R}}, \Gamma_{\mathbf{P}}) = (\mathbf{R}_1, \ldots, \mathbf{R}_{M^{CG}}, \mathbf{P}_1, \ldots, \mathbf{P}_{M^{CG}}) \ .$$

A general CG (hybrid) system may include particles corresponding to single "atoms", i.e. J for which $s_J = 1$, and into proper beads, i.e. J for which $s_J > 1$. The number of "atoms" in the CG system will be denoted as

$$N^A := \#\left\{J \in \left\{1, \ldots, M^{CG}\right\} : s_J = 1\right\} \ ,$$

while the number of beads is denoted as $N^B := M^{CG} - N^A$. For notational convenience, we will assume that the indices are ordered in such way that indices $J \in \left\{1, \ldots, N^A\right\}$ correspond to atoms, and $J \in \left\{N^A + 1, \ldots, M^{CG}\right\}$ corresponds to beads. A sketch of a possible mapping between FG and CG systems is shown in fig. 2.1.

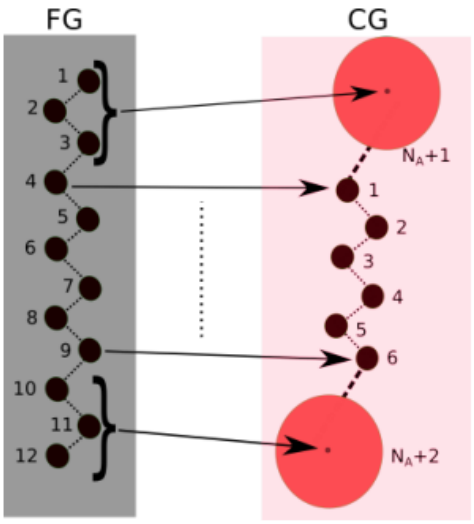


Figure 2.1: Sketch of the mapping of a molecule from the FG system to the CG system where the index mapping functions between the different resolutions are highlighted. In the picture, $N^A = 6$, $M^{CG} = 8$, therefore, $S_1 = \{4\}$, $S_2 = \{5\}$, ..., $S_6 = \{9\}$, $S_7 = \{1, 2, 3\}$, $S_8 = \{10, 11, 12\}$. [3]

2.2 A particular class of mapping operator

The mapping operators are a collection of functions defined in the FG system that specify the physical meaning of CG variables, i.e. they fix the CG variables will be observables of the FG system. Therefore they are functions

$$\Psi, \Phi \colon \mathbb{R}^{6M^{FG}} \to \mathbb{R}^{6M^{CG}}$$

such that

$$\boldsymbol{\gamma} \mapsto \Psi(\boldsymbol{\gamma}) =: \boldsymbol{\Gamma}_{\mathbf{R}}(\boldsymbol{\gamma})\,,$$
$$\boldsymbol{\gamma} \mapsto \Phi(\boldsymbol{\gamma}) =: \boldsymbol{\Gamma}_{\mathbf{P}}(\boldsymbol{\gamma})\,.$$

Chapter 3

Numerical Method

3.1 Langevin Dynamics Integrator: BAOAB method

In this work we simulated the Langevin Dynamics by the BAOAB method, which was derived and analysed by B. Leimkuhler and C. Matthews [12].

The aim of their work is to compare Langevin dynamics integrators in term of their invariant distributions and demonstrate a superconvergence property of a certain method in the high friction limit (or in the null mass limit). Numerically they implemented the various methods considered and compared the accuracy of configurational sampling for different values of the friction parameter γ and a range of timesteps.

Their results show that the best method to sample the configurational Gibbs-Boltzmann distribution[1] is the BAOAB method and in the context of molecular dynamics it is a cheap and easy to implement scheme using only a single force vector per timestep:

$$
\begin{aligned}
p_{n+1/2} &= p_n - \frac{1}{2}\nabla U\left(x_n\right)\, dt\,; \\
x_{n+1/2} &= x_n + \frac{1}{2}\, M^{-1}\, p_{n+1/2}\, dt\,; \\
\hat{p}_{n+1/2} &= c_1\, p_{n+1/2} + c_2\, M^{1/2}\, R_{n+1}\,; \\
x_{n+1} &= x_{n+1/2} + \frac{1}{2}\, M^{-1}\, \hat{p}_{n+1/2}\, dt\,; \\
p_{n+1} &= \hat{p}_{n+1/2} - \frac{1}{2}\nabla U\left(x_{n+1}\right)\, dt\,.
\end{aligned}
$$

Here, R_i is an N-vector of iid normal random numbers with mean 0 and variance 1, M is the diagonal mass matrix, dt is the timestep,

$$
c_1 = e^{-\tilde{\gamma}\, dt} \quad \text{and} \quad c_2 = \sqrt{\left(1 - c_1^2\right)/\beta}
$$

are evaluated for a given friction parameter $\tilde{\gamma} = \gamma\, M^{-1}$ and a given temperature parameter $\beta = (K_B T)^{-1}$, where K_B is the Boltzmann's constant.

[1] This means the calculation of functions averages of phase space variables of a molecular N-body system modelled at constant temperature.

3.1.1 An idea of how the BAOAB method was derived

An Itô stochastic differential equation (SDE) can be written as

$$d\mathbf{x}(t) = \mathbf{a}\,[\mathbf{x}(t), t]\,dt + \mathbf{b}\,[\mathbf{x}(t), t]\,d\mathbf{W}(t)\,,$$

where $\mathbf{x}$ are the n stochastic process, $\mathbf{a}(\mathbf{x}, t)$ and $\mathbf{b}(\mathbf{x}, t)$ are certain know functions, $d\mathbf{W} \equiv \boldsymbol{\xi}(t)\,dt$ is an n variable Wiener process and $\boldsymbol{\xi}(t)$ is the white noise. This SDE corresponds an Fokker-Planck equation if the stochastic process described by a conditional probability density $p(\mathbf{x}, t\,|\,\mathbf{x_0}, t_0) \equiv p$ satisfy

$$\partial_t\, p = \mathscr{L}(\mathbf{x}, t)\, p = -\sum_i \partial_i\,[A_i(\mathbf{x}, t)\, p\,] + \frac{1}{2}\sum_{ij} \partial_i\partial_j\,[B_{ij}(\mathbf{x}, t)\, p\,]$$

where $\mathscr{L}(\mathbf{x}, t)$ is a second-order differential operator and the coefficients of RHS are such that $\mathbf{A}(\mathbf{x}, t) = \mathbf{a}(\mathbf{x}, t)$ and $\underline{\mathbf{B}}(\mathbf{x}, t) = \mathbf{b}(\mathbf{x}, t)\mathbf{b}^T(\mathbf{x}, t)$. The Fokker-Planck equation describes diffusion processes, the vector $\mathbf{A}(\mathbf{x}, t)$ is know as drift vector and the matrix $\underline{\mathbf{B}}(\mathbf{x}, t)$ is know as the diffusion matrix. From its definition this matrix is positive semidefinite and symmetric [5].

Let us consider an Orstein-Uhlenbeck (OU) SDE, that is, an Itô equation where the coefficients are

$$\mathbf{a}(\mathbf{x}) = -k\,\mathbf{x} \quad \text{and} \quad b = \sqrt{D}$$

here k and D are positive parameters. The solution of one dimensional OU-SDE can be computed by the Itô's formula applied to $y = xe^{k\,t}$[5]. We get

$$x(t) = x_0(t_0)\, e^{-k(t-t_0)} + \sqrt{D}\int_{t_0}^{t} dW(t')e^{-k(t-t')}\,,$$

and the average and variance are respectively:

$$\mathbb{E}[x(t)] = \mathbb{E}[x_0(t_0)]\, e^{-k(t-t_0)}\,,$$
$$\mathbb{E}\left[(x(t) - \mathbb{E}[x(t)])^2\right] = \left[Var(x_0) - \frac{D}{2k}\right] e^{-2k(t-t_0)} + \frac{D}{2k}\,.$$

We can easily show in the stationary limit ($t \gg t_0$) and applying the equipartition theorem that the diffusive parameter D is equal to $(2k)/\beta$.

Langevin dynamics combines a conservative dynamic with the OU stochastic vector field as a forcing for the momentum, so we are assuming that the stochastic force is not affected by the potential:

$$M\frac{d^2\mathbf{x}}{dt^2} = -\boldsymbol{\nabla}U(\mathbf{x}) - \gamma\,\frac{d\mathbf{x}}{dt} + \sqrt{(2\gamma)/\beta}\,\boldsymbol{\xi}(t)$$

where we introduced physical quantities: M is the mass of the particle, $\mathbf{x}$ is the coordinate of position, γ is the friction parameter and $\beta = (K_B T)^{-1}$ is the temperature parameter. We can rewrite the equation as

$$\begin{cases} d\mathbf{x} = \mathbf{v}\,dt \\ M d\mathbf{v} = [-\boldsymbol{\nabla}U(\mathbf{x}) - \gamma\,\mathbf{v}]\;dt + \sqrt{(2\gamma)/\beta}\,d\mathbf{W} \end{cases}$$

and for the canonical coordinates $\{\mathbf{q}(t), \mathbf{p}(t)\}$ we get

$$\begin{cases} d\mathbf{q} = M^{-1}\,\mathbf{p}\,dt \\ d\mathbf{p} = [-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p}]\;dt + \sqrt{(2\tilde{\gamma})/\beta}\,M^{1/2} d\mathbf{W} \end{cases}$$

where $\tilde{\gamma} = \gamma/M$. From the system of two first-order equations can we compute the drift and diffusion coefficients

$$\begin{aligned} (A_{\mathbf{q}})_i &= M^{-1}\,p_i\,; & (B_{\mathbf{qq}})_{ij} &= (B_{\mathbf{pq}})_{ij} = 0\,; \\ (A_{\mathbf{p}})_i &= (-\boldsymbol{\nabla} U(\mathbf{q}) - \tilde{\gamma}\,\mathbf{p})_i\;; & (B_{\mathbf{pp}})_{ij} &= \frac{2\tilde{\gamma}}{\beta}\,M\,\delta_{ij}\,; \end{aligned}$$

and thus determine the Fokker-Planck operator for the Langevin dynamics

$$\begin{aligned} \mathscr{L}_{LD}\,\varphi = &-M^{-1}\sum_i p_i\,\partial_{q_i}\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\sum_i \partial_{p_i}\,\varphi \;+ \\ &+\;\tilde{\gamma}\sum_i \partial_{p_i}\,p_i\,\varphi \;+\; \tilde{\gamma}\sum_i p_i\,\partial_{p_i}\,\varphi \;+\; \tilde{\gamma}\,K_B T\sum_i M\,\partial_i^2\,\varphi = \\ &= -M^{-1}\mathbf{p}\cdot\boldsymbol{\nabla}_q\,\varphi \;+\; \boldsymbol{\nabla} U(\mathbf{q})\cdot\boldsymbol{\nabla}_p\,\varphi \;+\; \tilde{\gamma}\,[\,\boldsymbol{\nabla}_p(\mathbf{p}\,\varphi) + K_B T\boldsymbol{\Delta}_p\,\varphi\,]\;, \end{aligned}$$

where $\boldsymbol{\Delta}_p$ is the mass-weighted Laplacian in the momenta and φ is a test function [17].

Splitting methods

3.1.2 Some more details about the work of B. Leimkuhler and C. Matthews

The methods that are considered are splitting integrators that decompose the stochastic vector field which drives the Langevin Dynamics into simpler vector fields which can be solved exactly. Nevertheless, the composition method that results cannot be directly related to a stochastic differential equation, but nonetheless the invariant measure associated to the numerical method can be derived, using the Baker-Campbell-Hausdorff (BCH) expansion, as an asymptotic series in two parameters: the stepsize and the coefficient γ^{-1}. The authors stress that their approach focuses on the truncation error of the invariant distribution, thus the greatest benefit would be seen only when the statistical error is well controlled. Furthermore, they observe that the least biased scheme from the point of view of the error introduced in configurational sampling is also as efficient as the alternatives and the most robust with respect the variation of parameters (stepsize and γ^{-1}), thus they emphasise that there is no price for the improvement in accuracy.

Part III

Numerical Experiments

Appendices

Appendix A

Elements of Probability Theory

We introduce some basic definition and results from probability theory. This material can be found in [18].

Definition A.1 (**Sample Space**)**.** The set of all events, such as all the possible outcomes of an experiment, is called sample space and is denoted by Ω.

Definition A.2 (σ**-algebra**)**.** A collection $\mathscr{F}$ of subset of Ω is called σ-algebra on Ω if[1]

(i) $\Omega \in \mathscr{F}$;
(ii) $F \in \mathscr{F} \implies F^c := \Omega \setminus F \in \mathscr{F}$;
(iii) $F_n \in \mathscr{F}$, $n \in \mathbb{N} \implies \bigcup_{n\in\mathbb{N}} F_n \in \mathscr{F}$.

Intuitively, the σ-algebra $\mathscr{F}$ contains all the useful information that is available about the random experiment that we are performing. One of its elements is called a $\mathscr{F}$-measurable subset of Ω or *event*.

Definition A.3 (**Measurable Space**)**.** A pair $(\Omega, \mathscr{F})$, where Ω is a set and $\mathscr{F}$ is a σ-algebra on Ω, is called measurable space.

Definition A.4 (**Probability Measure**)**.** A probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$ is a function

$$\mathbb{P}\colon \mathscr{F} \to [0,1];\ F \mapsto \mathbb{P}(F);$$

satisfying

(i) $\mathbb{P}(\Omega) = 1$;
(ii) if $F_n \in \mathscr{F}$ with $F_n \bigcap F_m = \emptyset$, $n \neq m$ and $n, m \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} F_n\right) = \sum_{n\in\mathbb{N}} \mathbb{P}\left(F_n\right) .$$

.

[1] We use ':=' to signify 'is defined to equal', and the following convention:

$$\mathbb{N} := \{1, 2, 3, \dots\} \subseteq \{0, 1, 2, 3, \dots\} =: \mathbb{Z}^+ \quad \text{and} \quad \mathbb{R}^+ := [0, +\infty) .$$

Definition A.5 (**Probability Space**)**.** A probability space is a triple $(\Omega, \mathscr{F}, \mathbb{P})$ comprising a set Ω, a σ-algebra $\mathscr{F}$ of subsets of Ω and a probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathscr{F})$.

Since we are usually interested in the consequence of the outcomes of an experiment, we introduce the notion of random variable that is intuitively a function of the outcomes. We do it following the Williams [18], through the definition of measurable function from $(\Omega, \mathscr{F}, \mathbb{P})$ to $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$[2].

Definition A.6 (**$\mathscr{F}$-measurable function**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ and $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be two measurable spaces. The function $\psi \colon \Omega \to \mathbb{R}$ is called $\mathscr{F}$-measurable function if

$$\psi^{-1}(B) \coloneqq \{\omega \in \Omega \colon \psi(\omega) \in B\} \in \mathscr{F}, \quad \forall\, B \in \mathscr{B}(\mathbb{R})\,.$$

Definition A.7 (**Random Variable**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A random variable is a $\mathscr{F}$-measurable function, thus

$$X \colon \Omega \to \mathbb{R}\,, \quad X^{-1} \colon \mathscr{B}(\mathbb{R}) \to \mathscr{F}\,.$$

Every random variable, carried by some probability triple $(\Omega, \mathscr{F}, \mathbb{P})$, induces a probability measure $\mathbb{P}_X$ on $(\mathbb{R}, \mathscr{B}(\mathbb{R}))$:

$$\begin{aligned} \mathbb{P}_X \colon \mathscr{B}(\mathbb{R}) &\to [0,1] \\ B &\mapsto \mathbb{P}_X(B) \coloneqq \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B))\,. \end{aligned}$$

The measure $\mathbb{P}_X$ is called *distribution* (or sometimes the *law*) of X. We can now define the *distribution function* of X:

$$\begin{aligned} F_X \colon \mathbb{R} &\to [0,1] \\ x &\mapsto F_X(x) \coloneqq \mathbb{P}\left(\{\omega \in \Omega \colon X(\omega) \leq x\}\right) = \mathbb{P}\left(X \leq x\right)\,. \end{aligned}$$

This function F_X indicates the likelihood that the random variable X is in $(-\infty, x]$.

Definition A.8 (**Probability density function**)**.** Let X a random variable and $f : \mathbb{R} \to \mathbb{R}$ a function such that

$$f(x) \geq 0 \quad \forall\, x \in \mathbb{R} \qquad \text{and} \qquad \int_{\mathbb{R}} f(x)\,\mathrm{d}x = 1\,.$$

We say that X has a *probability density function* (pdf) f_X if

$$\mathbb{P}\left(X \in B\right) = \int_B f_X(x)\,\mathrm{d}x \qquad \forall\, B \in \mathscr{B}(\mathbb{R})$$

Definition A.9 (**Expected value**)**.** Let $\mathcal{L}^1\left(\Omega, \mathscr{F}, \mathbb{P}\right)$ be the space of integrable function, see [18, p. 54]. We define the *expectation* $\mathbb{E}\left[X\right]$ of a random variable $X \in \mathcal{L}^1\left(\Omega, \mathscr{F}, \mathbb{P}\right)$ by

$$\mathbb{E}\left[X\right] \coloneqq \int_\Omega X\,\mathrm{d}\mathbb{P} \equiv \int_\Omega X(\omega)\,\mathbb{P}(\mathrm{d}\omega) = \int_{\mathbb{R}} x\,\mathbb{P}_X(\mathrm{d}x)\,,$$

In the last equality we changed variable $x = X(\omega)$ and used the definition of distribution of X, that is $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$.

[2] We will use the notation $\mathscr{B}(\mathbb{R})$ to denote the Borel σ-algebra of $\mathbb{R}$, the smallest σ-algebra containing all open sets of $\mathbb{R}$.

Proposition A.10. Let $h : \mathbb{R} \to \mathbb{R}$ a Borel measurable function and $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ a random variable.
Then $h(X) \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ if and only if $h \in \mathcal{L}^1(\Omega, \mathscr{B}, \mathbb{P}_X)$ and then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\, \mathbb{P}_X(\mathrm{d}x)\,.$$

Furthermore, if f_X is a pdf of X and

$$\mathbb{E}[|h(X)|] < \infty \iff \int_{\mathbb{R}} |h(x)|\, f_X(x)\, \mathrm{d}x < \infty\,,$$

then

$$\mathbb{E}[h(X)] = \int_{\mathbb{R}} h(x)\, f_X(x)\, \mathrm{d}x\,.$$

Proof. See Williams [18, pp. 68-69]. □

We now give a central definition of modern probability, that is a fundamental theorem (Kolmogorov, 1933).

Theorem A.11 (**Conditional expectation**)**.** Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space, $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ a random variable and $\mathscr{G}$ a sub-σ-algebra of $\mathscr{F}$, $\mathscr{G} \subseteq \mathscr{F}$. Then there exists a random variable $Y \in \mathcal{L}^1(\Omega, \mathscr{G}, \mathbb{P})$, that is $\mathscr{G}$-measurable and $\mathbb{E}[|Y|] < \infty$, such that

$$\int_G Y \mathrm{d}\mathbb{P} = \int_G X \mathrm{d}\mathbb{P}\,, \quad \forall\, G \in \mathscr{G}\,.$$

Moreover, if $\tilde{Y}$ is another random variable that has the same properties as Y, then $\tilde{Y} = Y$ *a.s.* (almost surely), i.e. $\mathbb{P}(\tilde{Y} = Y) = 1$.
Therefore, Y with these properties is called **a version of the conditional expectation** $\mathbb{E}[X|\mathscr{G}]$ of X given $\mathscr{G}$, and we write $Y = \mathbb{E}[X|\mathscr{G}]$ *a.s.*

Proof. See Williams [18, pp. 85-86]. □

Generally one identifies several versions and speaks of the conditional expectation $\mathbb{E}[X|\mathscr{G}]$, but we should think about it "*a.s.*".

Intuitively, we have already said that the σ-algebra contains all the useful information that is available on an experiment, while a random variable is a function of its outcomes. In the case the only information we have available regarding which sample point ω has been chosen is the set of values $Z(\omega)$ for every $\mathscr{G}$-measurable random variable Z, i.e. the information available is given by the σ-algebra generated by Z, $\sigma(Z) = \mathscr{G}$. Then the random variable $Y(\omega) = \mathbb{E}[X|\mathscr{G}](\omega)$ is the expected value of $X(\omega)$ given this information. The "*a.s.*" ambiguity in the definition is something with which, in general, one must live.

Definition A.12 (**Conditional expectation as least-squares-best predictor**)**.** Let $X \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ be a random variable such that $\mathbb{E}[X^2] < \infty$. Then the conditional expectation $Y = \mathbb{E}[X|\mathscr{G}]$ is a version of the orthogonal projection of X onto $\mathcal{L}^2(\Omega, \mathscr{G}, \mathbb{P})$.
Hence, Y is the least-squares-best $\mathscr{G}$-measurable predictor of X: amongst all $\mathscr{G}$-measurable functions, i.e. amongst all predictors which can be computed from the available information, Y minimize $\mathbb{E}\left[(Y - X)^2\right]$ [18, p. 85].

In other words, this definition says that $Y = \mathbb{E}[X|\mathscr{G}]$ is a version of the orthogonal projection of $X \in \mathcal{L}^2(\Omega, \mathscr{F}, \mathbb{P})$ onto $\mathcal{L}^2(\Omega, \mathscr{G}, \mathbb{P})$ if

$$\mathbb{E}\left[(X-Y)^2\right] \leq \mathbb{E}\left[(X-Z)^2\right],$$

$\forall\, Z$ $\mathscr{G}$-measurable such that $\mathbb{E}\left[Z^2\right] < \infty$.
For a list of properties of the conditional expectation see [18, p. 88].

Proposition A.13. Let $X, Z \in \mathcal{L}^1(\Omega, \mathscr{F}, \mathbb{P})$ be two random variables which have a *joint probability density function* $f_{X,Z}(x,z)$. Then

$$f_Z(z) = \int_{\mathbb{R}} f_{X,Z}(x,z)\mathrm{d}x$$

acts as pdf for Z.
Define the *elementary conditional pdf* $f_{X|Z}$ of X given Z via:

$$f_{X|Z}(x|z) \coloneqq \begin{cases} f_{X,Z}(x,z)/f_Z(z) & \text{if } f_Z(z) \neq 0 \\ 0 & \text{otherwise}. \end{cases},$$

Let h be a Borel function on $\mathbb{R}$ such that $\mathbb{E}[|h(X)|] < \infty$. Set

$$g(z) \coloneqq \int_{\mathbb{R}} h(x) f_{X|Z}(x|z)\,\mathrm{d}x,$$

then $Y \coloneqq g(Z)$ is a version of the conditional expectation of $h(X)$ given $\sigma(Z)$, $\mathbb{E}[h(X)|\,\sigma(Z)] \equiv \mathbb{E}[h(X)|\,Z]$.

Proof. We can characterize $\sigma(Z)$ as

$$\sigma(Z) = \left\{A = Z^{-1}(B),\ \text{for } B \in \mathscr{B}(\mathbb{R})\right\} \subseteq \mathscr{F}$$

where

$$Z^{-1}(B) = \{\omega \in \Omega \colon Z(\omega) \in B\} \in \mathscr{F} \quad \forall\, B \in \mathscr{B}(\mathbb{R}).$$

We introduce the indicator function of A as

$$I_A(\omega) \coloneqq \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{otherwise}, \end{cases}$$

and from the characterization of $\sigma(Z)$ one has that $I_A(\omega) = I_B(Z)$, since that $\omega \in A$ if and only if $Z(\omega) \in B$.
Hence, to show that

$$g(Z) = \mathbb{E}[h(X)|\,Z] = \int_{\mathbb{R}} h(x) f_{X|Z}(x|z)\,\mathrm{d}x$$

for the theorem (A.11) we must show that

$$\mathbb{E}[g(Z)I_A] = \mathbb{E}[h(X)I_A] \quad \forall\, A \in \sigma(Z).$$

So we have, applying Fubini's theorem see [18, p. 77-79],

$$\begin{aligned}
&\mathbb{E}\left[g(Z)I_A\right] = \mathbb{E}\left[g(Z)I_B(Z)\right] = \int_B g(z)f_Z(z)\,\mathrm{d}z = \\
&= \int_{\mathbb{R}} g(z)I_B(z)f_Z(z)\,\mathrm{d}z = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} h(x) f_{X|Z}\left(x|z\right)\,\mathrm{d}x \right) I_B(z)f_Z(z)\,\mathrm{d}z = \\
&\quad = \int_{\mathbb{R}}\int_{\mathbb{R}} h(x)I_B(z)\, f_{X,Z}(x,z)\,\mathrm{d}x\,\mathrm{d}z = \mathbb{E}\left[h(X)I_B(Z)\right] = \mathbb{E}\left[h(X)I_A\right]\,.
\end{aligned}$$

□

Appendix B

Elements of Stochastic Processes

Appendix C

Liouville Space

In classical mechanics, the phase-space[1] $X = \mathbb{R}^{6N}$ is the space of all possible states of a physical system composed by N particles. The state is characterized by the instantaneous positions and momenta of the N particles of the system

$$\boldsymbol{\gamma} = \left(\boldsymbol{\gamma}_{\mathbf{r}}, \boldsymbol{\gamma}_{\mathbf{p}}\right) = (\mathbf{r}_1, \ldots, \mathbf{r}_N, \mathbf{p}_1, \ldots, \mathbf{p}_N) \; . \tag{C.1}$$

Let $\mathcal{H} : X \to \mathbb{R}$ the Hamiltonian of the system and we define a probability measure of finding the system in state $\boldsymbol{\gamma}$ as

$$\mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \rho(\boldsymbol{\gamma})\mathrm{d}\boldsymbol{\gamma} \, ,$$

where $\rho(\boldsymbol{\gamma})$ can be any equilibrium ensemble pdf. For example, in the canonical ensemble $\rho(\boldsymbol{\gamma}) = \mathcal{Z}^{-1} e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}$ where $\beta = (K_B T)^{-1} > 0$ such that $e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}$ is integrable with respect to the Lebesgue measure on X and $\mathcal{Z}$ is the canonical partition function, $\mathcal{Z} = \int_X e^{-\beta\mathcal{H}(\boldsymbol{\gamma})}\mathrm{d}\boldsymbol{\gamma}$.

The evolution of a state is obtained in according to the canonical equations of motion as

$$\begin{aligned} \frac{\mathrm{d}\boldsymbol{\gamma}}{\mathrm{d}t} &= \sum_{i=1}^{N} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \dot{\mathbf{r}}_i + \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \dot{\mathbf{p}}_i \right] = \\ &= \sum_{i=1}^{N} \left[\frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{r}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} - \frac{\partial \boldsymbol{\gamma}}{\partial \mathbf{p}_i} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \right] \equiv \{\boldsymbol{\gamma}, \mathcal{H}\} \end{aligned} \tag{C.2}$$

where $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is Poisson bracket of $\boldsymbol{\gamma}$ with the Hamiltonian $\mathcal{H}$. The equation obtained indicates that $\{\boldsymbol{\gamma}, \mathcal{H}\}$ is a generator of the time evolution of $\boldsymbol{\gamma}(t)$.

[1] Mathematically, the space of configuration might be defined by a *manifold* M (either finite or infinite dimensional), and for each position $\mathbf{r} \in M$ in that space, the momentum $\mathbf{p}$ of the system would take values in the cotangent space $T^*_{\mathbf{r}}M$ of that space. This since the velocity $\dot{\mathbf{r}}$ naturally lives in the tangent space $T_{\mathbf{r}}M$, but the momentum, for its definition, lives naturally in the cotangent space $T^*_{\mathbf{r}}M$:

$$\mathbf{p} \coloneqq \frac{\partial L(\mathbf{r}, \dot{\mathbf{r}})}{\partial \dot{\mathbf{r}}},$$

where $L(\mathbf{r}, \dot{\mathbf{r}})$ is the Lagrangian of the system. Therefore the phase-space is naturally represented by the *cotangent bundle*

$$T^*M \coloneqq \{(\mathbf{r}, \mathbf{p}) : \; \mathbf{r} \in M, \, \mathbf{p} \in T^*_{\mathbf{r}}M\} \; ,$$

which comes with a canonical *symplectic form* $\omega \coloneqq \mathrm{d}\mathbf{p} \wedge \mathrm{d}\mathbf{r}$.

The Poisson bracket allows us to introduce an operator L called *Liouvillian* or *Liouville operator*

$$iL \coloneqq \{\ldots, \mathcal{H}\} = \sum_{i=1}^{N} \left[\frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \cdot \frac{\partial}{\partial \mathbf{r}_i} - \frac{\partial \mathcal{H}}{\partial \mathbf{r}_i} \cdot \frac{\partial}{\partial \mathbf{p}_i} \right] \tag{C.3}$$

where $i = \sqrt{-1}$.

The equation (C.2) can be rewrite as $\mathrm{d}\boldsymbol{\gamma}/\mathrm{d}t = iL\,\boldsymbol{\gamma}$, which can be formally resolved as

$$\boldsymbol{\gamma}(t) = e^{iLt}\,\boldsymbol{\gamma}(0)\,,$$

where the operator $exp(iLt)$ is know as the *classical propagator* and defines a mapping in phase-space.

Let $A : X \to \mathbb{R}$ a *phase functions* satisfying $A \in L^2(X, \mathbb{P})$, where $L^2(X, \mathbb{P})$ is the Lebesgue space of square-summable functions and where functions are identified if and only if they differ only on a set of measure zero. If $L^2(X, \mathbb{P})$ comes with the usual scalar product

$$(A, B) = \int_X A(\boldsymbol{\gamma}) B(\boldsymbol{\gamma})\, \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}) = \int_X A(\boldsymbol{\gamma}) B(\boldsymbol{\gamma})\, \rho(\boldsymbol{\gamma})\, \mathrm{d}\boldsymbol{\gamma}\,, \tag{C.4}$$

then $L^2(X, \mathbb{P})$ is a Hilbert space[2]. It is simple to show that the rate of change of A is given by $\mathrm{d}A/\mathrm{d}t = iL\,A$, which formal solution is given by

$$A(\boldsymbol{\gamma}, t) = e^{iLt}\, A(\boldsymbol{\gamma}, 0)\,. \tag{C.5}$$

The Hilbert space just defined is often called *Liouville space* because the Liouvillian generates the motion in this space.

In a similar way to quantum mechanics, we can now show that the Liouvillian L is a linear Hermitian operator, that is $L^\dagger = L$ where $L^\dagger$ is Hermitian conjugate of L, and thereby the *classical propagator* is a unitary operator in the phase-space.

We can now define the *time correlation function* $C(t)$ as

$$C(t) \coloneqq \langle A(0)\, A(t) \rangle = \int_X A(\boldsymbol{\gamma}, 0) A(\boldsymbol{\gamma}, t)\, \rho(\boldsymbol{\gamma}) \mathrm{d}\boldsymbol{\gamma}\,, \tag{C.6}$$

where $\langle \ldots \rangle$ denotes an ensemble average. This equation can be expressed as

$$C(t) = \int_X A(\boldsymbol{\gamma})\, e^{iLt} A(\boldsymbol{\gamma})\, \rho(\boldsymbol{\gamma}) \mathrm{d}\boldsymbol{\gamma} = \big(A(\boldsymbol{\gamma}), e^{iLt} A(\boldsymbol{\gamma})\big) = (A(0), A(t))\,,$$

thus we obtained

$$\langle A(0)\, A(t) \rangle = (A(0), A(t))\,.$$

[2] In general, let X is a set and μ is a countably additive measure on the σ-algebra of subsets of X. The Hilbert space $L^2(X, \mu)$ is the space of complex-valued measurable functions such that

$$\int_X |f|^2 \mathrm{d}\mu < \infty,$$

where functions are identified if and only if they differ only on a set of measure zero and where the inner product of a pair of functions f, g in $L^2(X, \mu)$ is defined as

$$(f, g) = \int_X f^*\, g\, \mathrm{d}\mu.$$

Appendix D

Dunhamel-Dyson operator identity

The *resolvent* of an operator is define as its Laplace transform. We consider the resolvent of an exponential operator $exp(At)$:

$$R_A(s) = \int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-At}\,\mathrm{d}t = \int_0^\infty \mathbf{e}^{-st}\sum_{n=0}^\infty \frac{(-At)^n}{n!}\,\mathrm{d}t =$$
$$= \sum_{n=0}^\infty \frac{(-A)^n}{n!}\int_0^\infty \mathbf{e}^{-st}\,t^n\,\mathrm{d}t = \sum_{n=0}^\infty \frac{(-A)^n}{n!}\frac{1}{s^{n+1}}\int_0^\infty \mathbf{e}^{-st}\,(st)^n\,\mathrm{d}(st) =$$
$$= \sum_{n=0}^\infty \frac{(-A)^n}{n!}\frac{\Gamma(n+1)}{s^{n+1}} = \frac{1}{s}\sum_{n=0}^\infty \left(\frac{-A}{s}\right)^n = \frac{1}{s}\left(\mathcal{I}+\frac{A}{s}\right)^{-1} = (s+A)^{-1}\,,$$

where we used the definition of gamma function Γ and the fact that $\Gamma(n+1) = n!$. Therefore, the resolvent of the exponential operator of $-(A+B)t$ is given by

$$R_{A,B}(s) = (s+A+B)^{-1}\,.$$

We can rewrite the resolvent obtained through operator identity

$$(A+B)^{-1} = A^{-1} - A^{-1}B(A+B)^{-1},$$

which is easily verified by algebraic manipulations, we are of course assuming invertibility of the relevant operators. So we obtain

$$(s+A+B)^{-1} = (s+A)^{-1} - (s+A)^{-1}\,B\,(s+A+B)^{-1}\,.$$

Substituting the integral representation for the resolvent $R_A(s)$ and for the resolvent $R_{A,B}(s)$ into the equation obtained, we get

$$\int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-(A+B)t}\,\mathrm{d}t =$$
$$= \int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-At}\,\mathrm{d}t - \int_0^\infty \mathbf{e}^{-su}\mathbf{e}^{-Au}\,B\,\mathrm{d}u\int_0^\infty \mathbf{e}^{-sv}\mathbf{e}^{-(A+B)v}\,\mathrm{d}v =$$
$$= \int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-At}\,\mathrm{d}t - \int_0^\infty \mathbf{e}^{-Au}\,B\left[\int_0^\infty \mathbf{e}^{-s(u+v)}\mathbf{e}^{-(A+B)v}\,\mathrm{d}v\right]\mathrm{d}u\,.$$

Keeping u fixed and changing variable $t = u + v$, we can rewrite the internal integral in the equation to obtain

$$\int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-(A+B)t}\,\mathrm{d}t = \\ = \int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-At}\,\mathrm{d}t - \int_0^\infty \left[\int_u^\infty \mathbf{e}^{-Au}\,B\,\mathbf{e}^{-(A+B)(t-u)}\mathbf{e}^{-st}\,\mathrm{d}t\right]\mathrm{d}u\,.$$

The region of integration for the double integral is a triangular region in the (t, u) plane and we can reverse the order of integration in the integral to get

$$\int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-(A+B)t}\,\mathrm{d}t = \\ = \int_0^\infty \mathbf{e}^{-st}\mathbf{e}^{-At}\,\mathrm{d}t - \int_0^\infty \left[\int_0^t \mathbf{e}^{-Au}\,B\,\mathbf{e}^{-(A+B)(t-u)}\mathrm{d}u\right]\mathbf{e}^{-st}\,\mathrm{d}t = \\ = \int_0^\infty \mathbf{e}^{-st}\left[\mathbf{e}^{-At} - \int_0^t \mathbf{e}^{-Au}\,B\,\mathbf{e}^{-(A+B)(t-u)}\,\mathrm{d}u\right]\mathrm{d}t\,.$$

Since the equality holds for all values of s, the integrands must be equal, so:

$$\mathbf{e}^{-(A+B)t} = \mathbf{e}^{-At} - \int_0^t \mathbf{e}^{-Au}\,B\,\mathbf{e}^{-(A+B)(t-u)}\,\mathrm{d}u\,. \tag{D.1}$$

The derivation that we have used here is only valid if both of the operators A and B have no explicit time dependence. If we consider the operators $exp(-(A+B)t)$ and $exp(-At)$, then we get

$$\mathbf{e}^{(A+B)t} = \mathbf{e}^{At} + \int_0^t \mathbf{e}^{Au}\,B\,\mathbf{e}^{(A+B)(t-u)}\,\mathrm{d}u\,. \tag{D.2}$$

The equations obtained (D.1) and (D.2) are the Dyson operator identities, and when applied to propagators they are know as *Dyson decomposition* and *second Dyson decomposition* respectively [4].

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